

JACK BATES

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RESEARCH INTERESTS

Algorithmic accountability and the organisational consequences of automated decision systems: how retrieval- and scoring-based systems redistribute epistemic authority inside bureaucracies, who remains answerable when a judgement is produced by a model, and what makes such a decision reconstructable after the fact. Approached through a cybernetic frame — variety, control and feedback in the tradition of Wiener and Ashby — and grounded in the design and evaluation of systems actually deployed in an operational setting.

EDUCATION

University of Galway — BSc Business Information Systems 2023–2027

First Class Honours — Year 1 (71%), Year 2 (75%) · Academic Scholarship (Years 1 & 2)

- **Notable modules:** Advanced Database Technologies, Cybersecurity, Data Networks & Communications, Information Systems Technology, Advanced Application Development.

TBS Education, Barcelona — Bachelor of Management (Data Science for Business) 2025–2026

Double-degree exchange year.

- **Notable modules:** Programming, Big Data Modelling, Data Management & Visualization, The Business Aspect of AI, Econometrics, Data Governance.

Coláiste Choilm — Leaving Certificate 601 points

APPLIED RESEARCH & ENGINEERING EXPERIENCE

EUROCONTROL — Power Platform & AI Solutions Engineer Brussels · Apr–Aug 2026

Skyline Traineeship. Designed and deployed a suite of production AI agents for the Network Management Directorate.

- **Email triage & ticketing pipeline** — layered architecture over a shared service-desk mailbox: deterministic filters and request-type classification run first, an exception screen routes ambiguous mail to humans, and only clean cases reach the reasoning agent, which performs knowledge-base retrieval and returns a structured ticket / answer / route / discard decision. Evaluated over 25,000+ emails spanning ~15 months; measured ~40% of inbound volume as auto-ticketable and identified documentation coverage, not classification accuracy, as the binding constraint. Live in production.
- **Release Intelligence Agent** — ingests Microsoft 365 / Azure roadmap updates through a custom MCP server, assesses organisational impact against an internal knowledge base, and routes communications by priority band with human approval gating on high-impact items. Scoring is deliberately split from judgement: the model rates across seven weighted dimensions, deterministic code applies externally editable weights to compute the band, so any decision can be reconstructed and retuned without re-prompting. 800+ updates processed; ~8 hrs/week of manual monitoring displaced.
- **ITSD Help-Desk Agent** — retrieval assistant for service-desk staff, integrated with ServiceNow. Effort concentrated on constraining assertion rather than expanding retrieval: scope validation rejects near-miss articles with a stated reason, historical cases never substitute for documented procedure, and unanswerable queries escalate rather than produce a plausible guess.

Client Solution Architects — Shadowing Assistant. Digital solution implementation and client requirement workflows. 2020

PROJECTS

- **Writing-Voice LoRA Fine-Tune** — fine-tuned Llama-3.2-1B-Instruct (rank-32 LoRA, Unsloth) on 51 hand-curated examples of my own writing to test whether stylistic register transfers at small data scale, evaluated against the base model on held-out prompts. Style transferred; facts did not, and the fine-tune degraded below baseline out of distribution.
- **Healthcare Market Access Modelling (hackathon)** — classification, regression and clustering over 10,000+ observations across twelve European markets to model reimbursement approval and market share; applied multicollinearity diagnostics and class-imbalance handling. Found access conditions, not price, to be the dominant driver of adoption.
- **Personal Knowledge System** — agent-driven note system (Obsidian + Claude Code) turning raw notes and reading into a self-organising interlinked corpus — auto-tagging, cross-linking and flagging orphaned references.

SKILLS & CERTIFICATIONS

Technical: Python · SQL · R · Java · AI agent design (Copilot Studio, MCP connectors & tools) · LangGraph · Machine learning · MongoDB · Power BI

Certifications: ISO 8000 Master Data Quality Manager (ECCMA) · Enhance Agents with Autonomous Capabilities (Microsoft Applied Skills)

INTERESTS

Philosophy · Cybernetics · History

REFEREES

Available upon request.